

DESIGN OF COMPACT WIDEBAND SELF-ISOLATED MIMO ANTENNA FOR 5G MM WAVE USING SLOT DESIGN

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CERTIFICATE

This is to certify that the work entitled '**DESIGN OF COMPACT WIDEBAND SELF-ISOLATED MIMO ANTENNA FOR 5G mm WAVE USING SLOT DESIGN**' authored by G Sai Karthik, G Parvathi, and G Likitha Sri, has been carried out under my supervision. To the best of our knowledge, this work is original and has not been submitted elsewhere for any diploma or degree.

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DECLARATION

The work entitled '***DESIGN OF COMPACT WIDEBAND SELF-ISOLATED MIMO ANTENNA FOR 5G mm WAVE USING SLOT DESIGN***' which has been carried out in the Department of Electronics and Communication Engineering at Velagapudi Ramakrishna Siddhartha Engineering College, is original and conforms to the regulations of this Institute.

We understand the Institute's policy on plagiarism and declare that no part of this work has been copied from other sources or previously submitted elsewhere for any degree or diploma.

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Abstract

The growth of fifth-generation communication has increased the demand for compact MIMO antennas offering wide bandwidth and strong isolation at millimeter-wave frequencies. However, close element spacing in MIMO systems leads to significant mutual coupling, which degrades antenna performance. This work presents a compact self-isolated MIMO antenna using slot engineering on the radiating patch. U-shaped, H-shaped, and circular slots are introduced to control surface currents, enhance impedance bandwidth, and reduce mutual coupling between antenna elements. The antenna design is developed and analyzed using CST Studio Suite on a low-loss substrate. The proposed structure demonstrates wide impedance bandwidth, high isolation, and stable radiation characteristics without the need for external decoupling structures. Single, one-by-two, and two-by-two MIMO configurations are designed and evaluated, showing the effectiveness of the proposed self-isolation technique for 5G millimeter-wave applications.

Keywords: *5G communication, Millimeter-wave antenna, MIMO antenna, Self-isolated antenna, Mutual coupling reduction, Slot-based antenna design, CST Studio Suite*

Chapter 1

Introduction

1.1 Background

The rapid development of fifth-generation (5G) wireless communication systems has increased the demand for antennas capable of supporting high data rates, low latency, and reliable connectivity. To achieve these requirements, millimeter-wave (mm-wave) frequency bands in the range of 28–38 GHz are widely used, as they offer large available bandwidth. However, antenna design at mm-wave frequencies is challenging due to increased losses and strict size constraints in modern wireless devices.

Multiple-Input Multiple-Output (MIMO) technology is a key component of 5G systems, as it improves channel capacity, spectral efficiency, and communication reliability without increasing bandwidth or transmit power. In compact user equipment, multiple antenna elements must be placed very close to each other, which becomes unavoidable at mm-wave frequencies due to reduced physical dimensions.

When antenna elements are closely spaced, mutual coupling occurs due to electromagnetic interaction between adjacent radiators. Mutual coupling degrades antenna performance by affecting impedance matching, reducing isolation and radiation efficiency, and increasing the envelope correlation coefficient (ECC). These effects are more severe at mm-wave frequencies, where surface current interaction is stronger and antenna performance is highly sensitive to structural variations.

Several techniques such as electromagnetic band-gap (EBG) structures, metasurfaces, and neutralization lines have been proposed to reduce mutual coupling. Although effective, these methods often increase antenna size, require additional layers, and add design complexity. Therefore, there is a strong need for compact self-isolated MIMO antenna designs that achieve wide bandwidth and high isolation without external decoupling structures.

1.1.1 Fifth Generation (5G) Communication Systems

Fifth-generation (5G) wireless communication systems are designed to support extremely high data rates, ultra-low latency, and massive connectivity for modern applications such as autonomous vehicles, smart cities, and Internet of Things (IoT). To meet these requirements, 5G systems employ millimeter-wave (mm-wave) frequency bands, typically in the range of 28–38 GHz, where large bandwidths are available. The use of mm-wave frequencies enables multi-gigabit data transmission but also introduces challenges in antenna design due to increased propagation losses and strict size constraints.

1.1.2 MIMO Antennas for mm-Wave Applications

Multiple-Input Multiple-Output (MIMO) technology is widely used in 5G systems to enhance channel capacity, spectral efficiency, and link reliability without increasing bandwidth or transmit power. By using multiple antennas at the transmitter and receiver, MIMO systems effectively exploit multipath propagation. However, in compact mm-wave devices, antenna elements must be placed very close to each other due to limited available space, which makes the design of efficient MIMO antennas more challenging.

1.1.3 Mutual Coupling in Compact MIMO Antennas

Mutual coupling occurs when electromagnetic fields and surface currents from one antenna element interact with nearby elements. In compact MIMO antennas, this interaction leads to degraded impedance matching, reduced isolation, lower radiation efficiency, and increased envelope correlation coefficient (ECC). These effects significantly reduce the diversity performance of MIMO systems. At millimeter-wave frequencies, mutual coupling becomes more severe due to stronger surface wave propagation and higher sensitivity to antenna geometry.

1.1.4 Need for Self-Isolated MIMO Antenna Designs

Conventional isolation techniques such as electromagnetic band-gap structures, metasurfaces, and neutralization lines can reduce mutual coupling but often increase antenna size and design complexity. These limitations make them unsuitable for compact 5G devices. Hence, self-isolated MIMO antenna designs that provide wide bandwidth and high isolation without external decoupling structures are required. The proposed slot-based design addresses this need by effectively suppressing surface currents.

1.1.5 Challenges in Millimeter-Wave Antenna Integration

The integration of antennas operating at millimeter-wave (mm-wave) frequencies into compact wireless devices presents several technical challenges. Although mm-wave bands provide large available bandwidth, their practical implementation is constrained by high propagation losses, sensitivity to blockage, and stringent size limitations. At higher frequencies, free-space path loss increases significantly, requiring antennas with sufficient gain and efficient radiation characteristics to maintain reliable communication links.

Another major challenge in mm-wave antenna integration is the compactness of modern user equipment such as smartphones, tablets, and IoT devices. The reduced wavelength at mm-wave frequencies allows antennas to be physically smaller; however, the limited available space forces multiple antenna elements to be placed very close to each other. This close spacing leads to strong electromagnetic interaction between antenna elements, resulting in mutual coupling, impedance mismatch, and degradation of radiation performance.

Manufacturing tolerances also play a critical role at mm-wave frequencies. Small variations in antenna dimensions, substrate properties, or slot geometry can cause noticeable shifts in resonant frequency and bandwidth. Additionally, the selection of suitable substrate materials is challenging, as mm-wave antennas require low dielectric constant and low loss tangent substrates to minimize dielectric losses and maintain high efficiency.

Thermal effects and integration with other radio-frequency (RF) components further complicate mm-wave antenna design. Antennas must coexist with active circuits, packaging structures, and ground planes without significant performance degradation. These challenges collectively highlight the need for compact, robust, and efficiently isolated antenna designs that can operate reliably in practical mm-wave 5G environments.

1.1.6 Role of Isolation in MIMO Capacity Enhancement

Isolation between antenna elements is a critical factor that directly influences the performance of Multiple-Input Multiple-Output (MIMO) communication systems. In MIMO technology, multiple antennas are used to transmit and receive independent data streams simultaneously, thereby increasing channel capacity and spectral efficiency. However, the benefits of MIMO systems can only be fully realized when the antenna elements exhibit low mutual coupling and high isolation.

Poor isolation leads to increased signal correlation between antenna elements, which reduces spatial diversity and degrades MIMO capacity. High mutual coupling causes power leakage from one antenna element to another, resulting in reduced radiation efficiency and distorted radiation patterns. This effect increases the envelope correlation coefficient

(ECC), indicating that the received signals are highly correlated and thus provide limited diversity gain.

In compact mm-wave MIMO systems, achieving high isolation is particularly challenging due to the close proximity of antenna elements. Traditional isolation enhancement techniques such as electromagnetic band-gap structures, metasurfaces, and neutralization lines can improve isolation, but they often increase antenna size and design complexity. These limitations make them unsuitable for compact 5G devices.

Self-isolated antenna designs address this issue by intrinsically reducing coupling through optimized radiator geometry. By controlling surface current distribution using slot-based techniques, high isolation can be achieved without external decoupling structures. Improved isolation leads to lower ECC values, enhanced MIMO diversity performance, and increased channel capacity, making self-isolated MIMO antennas highly attractive for next-generation 5G mm-wave communication systems.

1.2 Motivation and Inspiration

The rapid growth of data-intensive applications such as ultra-high-definition video streaming, virtual and augmented reality, autonomous vehicles, and smart city infrastructure has significantly increased the demand for high-speed and reliable wireless communication systems. Fifth-generation (5G) networks aim to address these demands by utilizing mm-wave frequency bands and advanced MIMO technologies to achieve multi-gigabit data rates and ultra-low latency.

Despite the advantages of mm-wave communication, the practical realization of compact MIMO antennas remains a major challenge due to mutual coupling and limited device size. Many existing isolation techniques rely on additional structures such as EBGs or metasurfaces, which increase antenna footprint and fabrication complexity. This is not desirable for compact consumer devices where space and cost are critical constraints.

The motivation behind this work is to develop a compact and efficient MIMO antenna that achieves wide bandwidth and high isolation without relying on external decoupling structures. Slot-based antenna engineering provides a simple yet effective approach to modify surface current paths, enhance impedance bandwidth, and suppress mutual coupling. Recent studies have demonstrated the potential of slot-based designs; however, further exploration is required to optimize such techniques for compact multi-element mm-wave MIMO configurations.

This project is inspired by the need for practical, low-profile, and scalable antenna solutions that can be easily integrated into real-world 5G devices. By focusing on self-isolated

antenna geometry, the proposed work aims to contribute toward efficient antenna designs suitable for future wireless communication systems.

1.3 Research Gap Identification

From the literature review, it is evident that significant research has been conducted on improving bandwidth and isolation in MIMO antennas for 5G mm-wave applications. Techniques such as electromagnetic band-gap structures, metasurfaces, and neutralization lines have demonstrated effective coupling reduction; however, these approaches often increase antenna size, require multilayer fabrication, or introduce design complexity.

Slot-based self-isolated antenna designs have emerged as a promising alternative due to their compactness and simplicity. Nevertheless, many existing works focus on limited configurations or do not systematically analyze scalability from single-element designs to multi-element MIMO arrays. Additionally, there is limited investigation into the combined use of multiple slot geometries for simultaneous bandwidth enhancement and isolation improvement in compact mm-wave MIMO systems.

Therefore, a clear research gap exists in developing a compact, single-layer, slot-based self-isolated MIMO antenna that provides wide bandwidth, high isolation, and good diversity performance across different MIMO configurations. The present work addresses this gap by proposing and analyzing a compact antenna design employing U-shaped, H-shaped, and circular slots for single, 1×2 , and 2×2 MIMO configurations suitable for 5G millimeter-wave applications.

1.4 Objectives

The objective of this project is to design a compact microstrip antenna suitable for 5G millimeter-wave applications while addressing the issue of mutual coupling in closely spaced MIMO systems. The work aims to achieve wide impedance bandwidth and high isolation by employing slot-based self-isolation techniques without the use of external decoupling structures. In addition, the performance of single, 1×2 , and 2×2 MIMO antenna configurations is analyzed to evaluate improvements in isolation, gain, and efficiency. The antenna design and performance are validated through electromagnetic simulations using CST Studio Suite.

1.5 Scope

The scope of this work is limited to the design and simulation-based analysis of compact self-isolated MIMO antennas for 5G millimeter-wave applications. The study focuses on slot-based techniques to reduce mutual coupling and enhance bandwidth using single, 1×2 , and 2×2 MIMO configurations. All antenna designs are modeled and analyzed using CST Studio Suite, and performance parameters such as S-parameters, isolation, radiation characteristics, gain, and efficiency are evaluated. Fabrication and experimental validation are outside the scope of the present work and may be considered in future studies.

1.6 Thesis Organization

This thesis is structured into five chapters, each with a specific focus. A brief summary of these chapters is presented below.

Chapter 1 introduces the research topic and presents the background, motivation, objectives, scope, and overall structure of the thesis.

Chapter 2 reviews the fundamental concepts of 5G millimeter-wave communication, MIMO antenna systems, and existing techniques for mutual coupling reduction.

Chapter 3 describes the proposed antenna design methodology, including slot-based self-isolation techniques and MIMO configurations.

Chapter 4 presents the simulation results and performance analysis of the single, 1×2 , and 2×2 MIMO antennas.

Chapter 5 concludes the thesis by summarizing the work carried out and suggesting possible directions for future research.

Chapter 2

Literature Review

A literature review provides an overview of existing research related to the proposed work and helps identify research gaps. In the context of 5G millimeter-wave MIMO antennas, several techniques have been proposed to improve bandwidth and reduce mutual coupling between closely spaced antenna elements. This chapter reviews relevant studies on MIMO antenna designs, mutual coupling reduction techniques, and slot-based self-isolated antenna structures.

2.1 Background and Fundamental Concepts

This section presents the essential technical concepts required for the design and analysis of compact self-isolated MIMO antennas operating at millimeter-wave frequencies. Unlike the introductory discussion in Chapter 1, this section focuses on antenna-related fundamentals that directly support the proposed design methodology and performance evaluation.

2.1.1 Millimeter-Wave Propagation Characteristics

Millimeter-wave frequencies, typically ranging from 28 to 38 GHz for 5G applications, offer large bandwidths that enable high data rate communication. However, signal propagation at these frequencies is affected by increased free-space path loss, surface wave effects, and sensitivity to physical obstructions. These characteristics impose strict requirements on antenna design, including compact size, efficient radiation, and stable impedance behavior.

2.1.2 MIMO Antenna Systems

Multiple-Input Multiple-Output (MIMO) antenna systems employ multiple radiating elements to improve channel capacity, spectral efficiency, and communication reliability. In millimeter-wave systems, MIMO antennas are integrated into compact platforms, which necessitates close spacing between elements. This compact arrangement enhances spatial diversity but also increases the likelihood of electromagnetic interaction between antenna elements.

2.1.3 Mutual Coupling Mechanisms in MIMO Antennas

Mutual coupling in MIMO antennas arises due to near-field interactions and surface current coupling between closely spaced radiating elements. This phenomenon leads to degradation in impedance matching, reduced isolation, lower radiation efficiency, and increased envelope correlation coefficient (ECC). Understanding the mechanisms of mutual coupling is essential for designing effective isolation techniques, particularly in compact millimeter-wave MIMO configurations.

2.1.4 Slot-Based Surface Current Control

Slot-based techniques are widely used to control surface current distribution on microstrip antennas. By introducing slots such as U-shaped, H-shaped, and circular slots on the radiating patch, the effective current path length can be modified, resulting in multiple resonances and improved impedance bandwidth. Additionally, these slots help suppress coupling currents between adjacent elements, enabling self-isolation without the need for external decoupling structures. This concept forms the basis for the proposed compact MIMO antenna design.

2.2 Related Works- Review of Mutual Coupling Reduction Techniques

Several techniques have been proposed in the literature to mitigate mutual coupling in compact MIMO antenna systems, particularly for 5G millimeter-wave applications. These methods aim to improve isolation, reduce signal correlation, and enhance overall MIMO performance. The most widely reported approaches include electromagnetic band-gap (EBG) structures, metasurface-based isolation techniques, neutralization lines, and slot-based self-

isolated antenna designs. Each technique offers specific advantages and limitations, especially when applied to compact mm-wave antenna systems.

2.2.1 EBG-Based Isolation Techniques

Electromagnetic band-gap (EBG) structures are widely used to suppress surface waves and reduce mutual coupling between closely spaced antenna elements. Al-Hasan et al. demonstrated a compact EBG structure for millimeter-wave applications that significantly improved isolation between antenna elements [1]. Although EBG-based designs are effective, they often require additional layers or periodic structures, which increase antenna size and fabrication complexity, making them less suitable for compact mm-wave devices.

2.2.2 Metasurface-Based Isolation Techniques

Metasurface-based techniques have gained attention due to their strong ability to manipulate electromagnetic waves. Gupta et al. proposed metasurface corrugations to reduce mutual coupling in a 28 GHz MIMO antenna system [2]. The proposed design achieved high isolation levels; however, the complexity of metasurface fabrication and increased structural thickness limit its practical implementation in low-profile and cost-sensitive 5G devices.

2.2.3 Neutralization Line Techniques

Neutralization lines and near-field resonators provide isolation by introducing coupling paths that cancel undesired electromagnetic interaction. Li et al. employed near-field resonators as coupling mode transducers to enhance isolation in compact MIMO patch antennas [3]. While effective, such techniques often increase antenna footprint and may adversely affect impedance bandwidth, especially at millimeter-wave frequencies.

2.2.4 Slot-Based Self-Isolated Antenna Techniques

Slot-based techniques offer a compact and efficient approach to mutual coupling reduction by modifying surface current distribution on the radiating patch. Sokunbi et al. introduced slotted structures to achieve reduced mutual coupling in microstrip antenna arrays for 5G applications [4]. Slot-based self-isolated designs provide wide bandwidth, high isolation, and stable radiation characteristics without the need for external decoupling structures, making them highly suitable for compact millimeter-wave MIMO systems.

2.3 Limitations of Existing Approaches for Compact mmWave MIMO

Although existing techniques demonstrate effective mutual coupling reduction, their application in compact mm-wave MIMO systems remains limited. EBG and metasurface-based designs increase antenna size and fabrication complexity, which is undesirable for compact 5G devices. Neutralization lines, while simple, suffer from narrow bandwidth and sensitivity to dimensional variations.

In contrast, slot-based self-isolated designs provide intrinsic isolation without additional structures, maintaining compact size and wideband performance. However, many reported works focus on limited MIMO configurations or single slot geometries, leaving scope for further optimization and scalability analysis.

2.4 Problem Statements

Compact MIMO antenna design for 5G millimeter-wave applications is challenged by strong mutual coupling caused by closely spaced antenna elements. At millimeter-wave frequencies, limited device size leads to degradation in impedance bandwidth, isolation, radiation efficiency, and MIMO diversity performance. Although several isolation enhancement techniques exist, many rely on external decoupling structures that increase antenna size and design complexity. Therefore, there is a need for a compact self-isolated MIMO antenna design that achieves wide bandwidth and high isolation without external decoupling structures. This work addresses this problem by investigating a slot-based self-isolated MIMO antenna for single, 1×2 , and 2×2 configurations suitable for 5G millimeter-wave applications.

2.5 Proposed Solution

To overcome the problem of mutual coupling in compact MIMO antennas for 5G millimeter-wave applications, a slot-based self-isolated antenna design is proposed. The solution employs U-shaped, H-shaped, and circular slots etched on the radiating patch to modify surface current distribution and suppress coupling between closely spaced antenna elements, without external decoupling structures.

The proposed antenna is implemented using a microstrip patch configuration and analyzed through electromagnetic simulation. Single, 1×2 , and 2×2 MIMO configurations are studied

to evaluate the effectiveness of the slot-based approach in achieving wide bandwidth, high isolation, and stable radiation characteristics suitable for 5G millimeter-wave systems.

Chapter 3

Proposed Approach

This chapter describes the methodology adopted for the design and analysis of a compact self-isolated MIMO antenna for 5G millimeter-wave applications. The proposed approach is systematic and reproducible, beginning with antenna design and progressing through slot optimization, MIMO configuration, and performance evaluation. The methodology is presented in a step-by-step manner to clearly explain the design choices, tools used, and analysis techniques employed. A visual representation of the overall methodology is shown in Fig. 3.1.

3.1 Detailed Design Methodology

The design of the proposed compact self-isolated MIMO antenna follows a systematic and structured methodology to ensure wide bandwidth, high isolation, and compact size suitable for 5G millimeter-wave applications. The methodology begins with the design of a conventional microstrip patch antenna operating in the desired frequency band and progressively incorporates slot-based modifications to enhance performance. The optimized single antenna element is then extended to multi-element MIMO configurations to evaluate isolation and diversity characteristics.

Each stage of the design process is validated through electromagnetic simulation using CST Studio Suite. The step-by-step approach ensures repeatability, ease of optimization, and practical feasibility of the proposed antenna design.

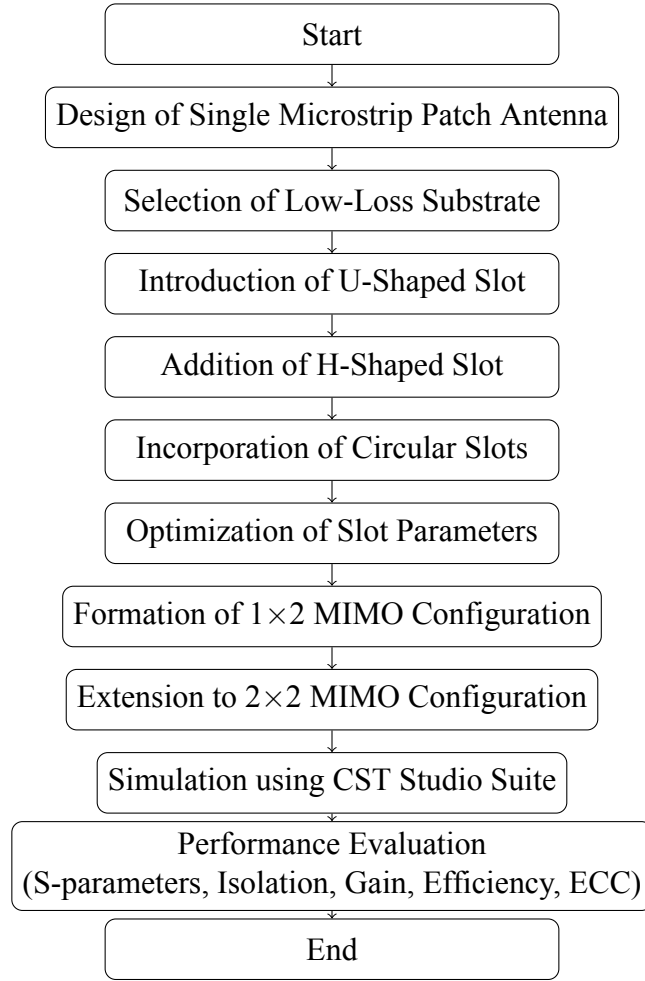


Figure 3.1: Proposed methodology for slot-based self-isolated MIMO antenna design

3.2 Mathematical Design of Microstrip Patch Antenna

The initial dimensions of the rectangular microstrip patch antenna are calculated using standard transmission-line model equations. These equations provide an approximate starting point, which is further optimized through simulation to account for fringing effects and substrate properties.

3.2.1 Patch Width Calculation

The width of the rectangular patch is given by:

$$W = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}}$$

where:

- c is the velocity of light in free space,
- f_r is the resonant frequency of the antenna,
- ε_r is the dielectric constant of the substrate material.

A wider patch improves radiation efficiency and impedance bandwidth, which is particularly beneficial for millimeter-wave antenna applications.

3.3 Slot-Based Self-Isolation Technique

To enhance bandwidth and reduce mutual coupling, slot-based self-isolation techniques are employed on the radiating patch. U-shaped and H-shaped slots are introduced to alter the surface current distribution and generate multiple resonant modes. These slots increase the effective current path length, thereby improving impedance bandwidth.

In addition, circular slots are incorporated on the patch to further suppress surface currents responsible for coupling between closely spaced antenna elements. The slot dimensions, orientation, and placement are optimized through parametric analysis to achieve effective self-isolation without using external decoupling structures. This approach enables a compact and single-layer antenna design suitable for 5G millimeter-wave systems.

3.4 MIMO Configuration Development

After optimizing the single antenna element, the design is extended to MIMO configurations to evaluate isolation and diversity performance. A 1×2 MIMO configuration is formed by placing two identical antenna elements with minimal inter-element spacing while maintaining compactness. The optimized slot structure helps reduce mutual coupling between the antenna elements in this configuration.

Further, a 2×2 MIMO configuration is developed to study scalability and multi-element performance. The antenna elements are arranged symmetrically, and the same slot-based self-isolation technique is applied to all elements. This configuration allows evaluation of the proposed design for practical multi-antenna 5G applications.

3.5 Simulation and Performance Evaluation

All antenna designs are modeled and analyzed using CST Studio Suite. Parametric sweeps are performed to optimize key design parameters such as slot dimensions and feed position.

The antenna performance is evaluated in terms of reflection coefficient (S-parameters), isolation between antenna elements, radiation patterns, gain, efficiency, and envelope correlation coefficient (ECC).

The simulation results obtained for the single, 1×2 , and 2×2 MIMO configurations are used to verify the effectiveness of the proposed slot-based self-isolated antenna design. Detailed analysis of these results is presented in the subsequent chapter.

Chapter 4

Results and Discussion

This chapter presents the simulation results and performance analysis of the proposed compact self-isolated MIMO antenna designed for 5G millimeter-wave applications. The results are obtained using CST Studio Suite and are discussed for different antenna configurations, including the single-element antenna, 1×2 MIMO antenna, and 2×2 MIMO antenna. Key performance parameters such as S-parameters and far-field radiation characteristics are analyzed to evaluate the effectiveness of the proposed slot-based self-isolation technique

4.1 Antenna Geometry and Configurations

The geometry of the proposed antenna includes a compact microstrip patch with U-shaped, H-shaped, and circular slots etched on the radiating element. The antenna is implemented on a Rogers RT/Duroid 5870 substrate with a copper ground plane. The single antenna element is extended to form 1×2 and 2×2 MIMO configurations while maintaining compact dimensions. The layouts of the ground, substrate, patch, and MIMO configurations are shown in Fig. 4.1

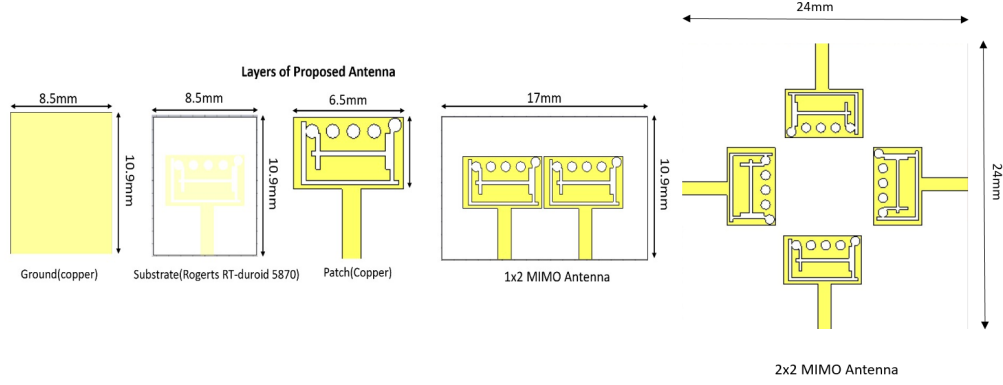


Figure 4.1: Geometry and configurations of the proposed single element, 1x2, and 2x2 MIMO antennas

4.2 S-Parameter Analysis

The simulated S-parameters of the proposed MIMO antenna configurations are illustrated in Fig. 4.2 and Fig. 4.3. For the 1x2 MIMO antenna, the reflection coefficient (S_{11}) remains below -10 dB over the desired 5G millimeter-wave frequency band, indicating good impedance matching. The transmission coefficients (S_{11} and S_{21}) demonstrate high isolation between the antenna elements, confirming the effectiveness of the slot-based self-isolation approach.

In the 2x2 MIMO configuration, all reflection coefficients exhibit satisfactory impedance matching across the operating band. The mutual coupling between adjacent antenna elements is significantly reduced, with isolation levels remaining well below -20 dB across most of the band. These results validate that the proposed design achieves strong isolation without the use of external decoupling structures.

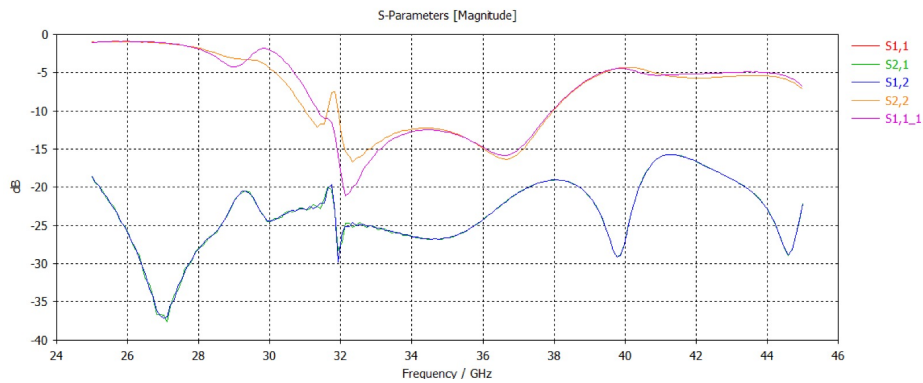
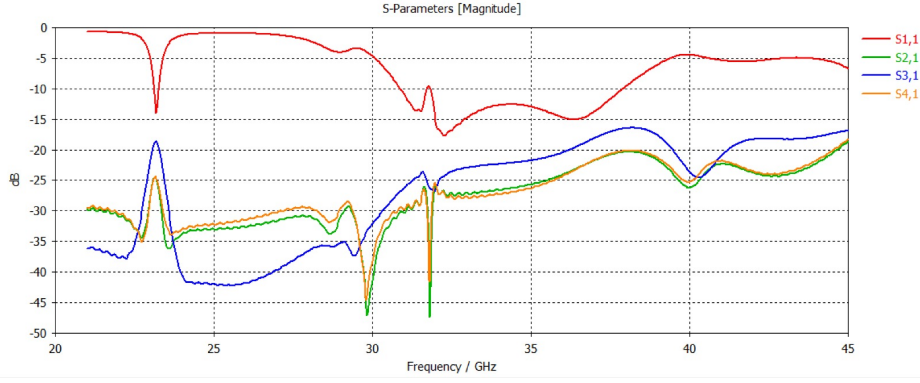


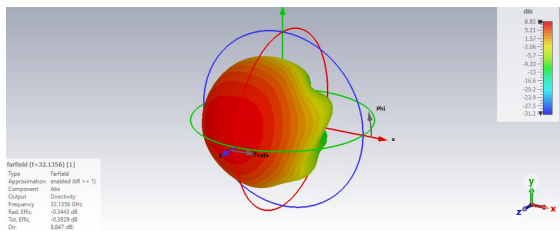
Figure 4.2: Simulated S-parameters of the proposed 1x2 MIMO antenna

Figure 4.3: Simulated S-parameters of the proposed 2×2 MIMO antenna

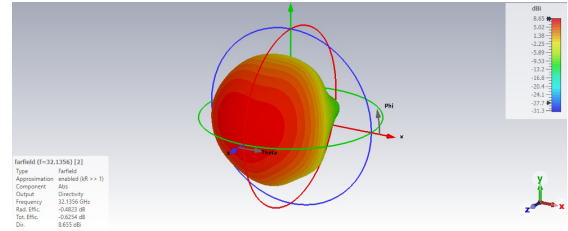
4.3 Far-Field Radiation Characteristics

The three-dimensional far-field radiation patterns of the proposed antenna are presented for both 1×2 and 2×2 MIMO configurations at representative operating frequencies. The radiation patterns show stable and directional behavior with well-defined main lobes. The antenna exhibits good directivity and consistent radiation characteristics across the operating band.

The far-field plots indicate that the introduction of slots does not distort the radiation pattern significantly. Similar radiation behavior is observed among different antenna elements, which is desirable for MIMO systems. The results confirm that the proposed antenna design maintains acceptable radiation performance while achieving compact size and high isolation.



(a) Far-field radiation pattern at 32.1 GHz (Port1)



(b) Far-field radiation pattern at 32.1 GHz (Port2)

Figure 4.4: 3D far-field radiation patterns of the proposed 1×2 MIMO antenna

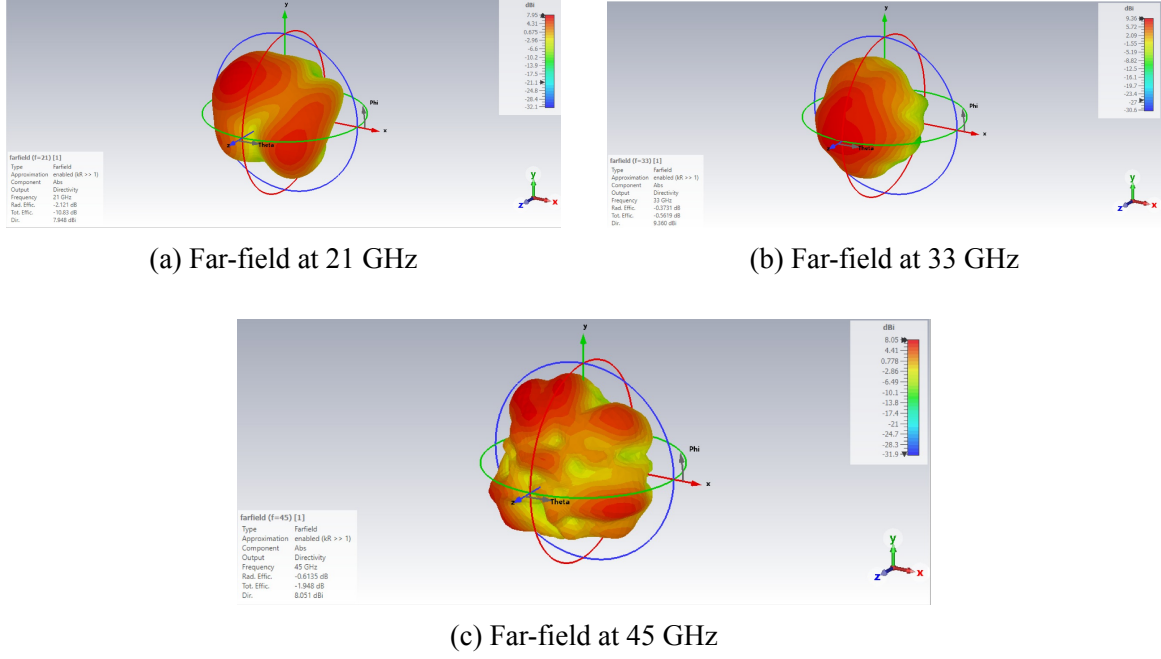


Figure 4.5: 3D far-field radiation patterns of the proposed 2×2 MIMO antenna

4.4 Performance Comparison and Discussion

Table 4.1 presents a comparative performance analysis of the proposed 1×2 and 2×2 MIMO antenna configurations in terms of impedance bandwidth, isolation, gain, and radiation efficiency. The comparison clearly demonstrates the effectiveness of the slot-based self-isolation technique implemented in the proposed antenna design.

The 1×2 MIMO antenna achieves an impedance bandwidth of 4.4594 GHz, while the 2×2 MIMO configuration exhibits a wider bandwidth of 5.479 GHz. The increase in bandwidth for the 2×2 configuration is attributed to enhanced coupling of multiple resonant modes introduced by the optimized slot structures and array configuration.

In terms of isolation performance, the 1×2 MIMO antenna provides an isolation level of approximately −25 dB at 32 GHz, indicating adequate suppression of mutual coupling between the antenna elements. The 2×2 MIMO antenna exhibits significantly improved isolation of about −45 dB at 32 GHz, which highlights the effectiveness of symmetric slot placement and orthogonal element orientation in reducing surface current interaction.

Regarding radiation characteristics, the 1×2 MIMO antenna achieves a peak gain ranging from 8.65 dBi to 8.85 dBi, whereas the 2×2 MIMO configuration attains a higher gain of 9.36 dBi due to array effects. Radiation efficiency remains high for both configurations, with efficiencies of 86.8

Overall, the results validate that the proposed slot-engineered MIMO antenna design suc-

cessfully achieves a balanced trade-off among compact size, wide bandwidth, high isolation, and good radiation performance, making it suitable for 5G millimeter-wave communication applications.

Table 4.1: Performance comparison of the proposed 1×2 and 2×2 MIMO antennas

Parameter	1×2 MIMO Antenna	2×2 MIMO Antenna
Bandwidth (GHz)	4.4594	5.479
Isolation (dB)	-25 at 32 GHz	-45 at 32 GHz
Gain (dBi)	8.65–8.85	9.36
Radiation Efficiency (%)	86.8	87.89

4.5 Cause–Effect Analysis of Slot-Based Improvements

The observed enhancements in bandwidth, isolation, and gain are primarily attributed to the incorporation and optimization of slot structures on the radiating patches. The U-shaped and H-shaped slots increase the effective surface current path length, leading to the excitation of multiple resonant modes and resulting in improved impedance bandwidth.

The circular slots play a crucial role in reducing mutual coupling by suppressing surface currents responsible for near-field interaction between adjacent antenna elements. These slots create regions of current cancellation, thereby significantly reducing electromagnetic coupling without degrading the primary radiation characteristics.

The substantial improvement in isolation from -25 dB in the 1×2 configuration to -45 dB in the 2×2 configuration is attributed to symmetric slot placement, orthogonal orientation of antenna elements, and enhanced surface current cancellation mechanisms. Additionally, the array configuration in the 2×2 MIMO antenna contributes to higher gain through constructive field superposition.

As a result, the proposed 2×2 MIMO antenna achieves both enhanced isolation and increased gain, which are typically conflicting requirements in compact MIMO systems. These outcomes confirm the effectiveness of the proposed slot-based self-isolation technique for high-performance 5G millimeter-wave MIMO antenna applications.

4.6 Performance Metrics: Envelope Correlation Coefficient (ECC) and Total Active Reflection Coefficient (TARC)

The envelope correlation coefficient (ECC) is a key parameter used to evaluate the diversity performance of MIMO antenna systems. Physically, ECC represents the degree of corre-

lation between signals received by different antenna elements. Lower ECC values indicate better diversity performance and reduced signal correlation, which are essential for reliable MIMO communication.

In this work, ECC is calculated using the S-parameters of the antenna system and is expressed as:

$$\text{ECC} = \frac{|S_{11}S_{12}^* + S_{21}S_{22}^*|^2}{(1 - |S_{11}|^2 - |S_{21}|^2)(1 - |S_{22}|^2 - |S_{12}|^2)} \quad (4.1)$$

For practical MIMO systems, ECC values less than 0.5 are considered acceptable, while values below 0.05 indicate excellent diversity performance. The proposed slot-based self-isolated MIMO antenna exhibits very low ECC values due to effective suppression of mutual coupling and reduced surface current interaction between antenna elements.

The total active reflection coefficient (TARC) is another important metric used to assess the impedance behavior of MIMO antennas under simultaneous multi-port excitation. Unlike the reflection coefficient (S11), which considers single-port excitation, TARC accounts for the combined reflected power when multiple ports are excited with a phase difference.

TARC is given by:

$$\text{TARC} = \frac{|S_{11} + S_{12}e^{j\theta}|^2 + |S_{21} + S_{22}e^{j\theta}|^2}{2} \quad (4.2)$$

Here, θ represents the phase difference between the excitation signals at different antenna ports. Stable and low TARC values indicate good impedance matching and efficient radiation under multi-port excitation conditions.

The obtained low ECC and stable TARC values confirm that the proposed antenna provides good MIMO diversity performance and is well suited for 5G millimeter-wave communication applications.

4.7 Parametric Study of Slot Dimensions

A parametric study was carried out to optimize the dimensions and placement of the slot structures on the radiating patch. Increasing the slot length was observed to shift the resonant frequency toward lower values due to the increase in effective current path length. This behavior was utilized to achieve wide impedance bandwidth.

Variation in slot width significantly influenced isolation performance. Wider slots enhanced coupling suppression by disrupting surface current flow; however, excessive widening led to impedance mismatch. Hence, an optimal slot width was selected to balance bandwidth and isolation.

4.7 Parametric Study of Slot Dimensions

The position of the slots relative to high current density regions on the patch also played a crucial role. Slots placed closer to these regions provided stronger current cancellation and improved isolation. Based on these observations, the final slot configuration was optimized to achieve high isolation, stable radiation patterns, and good overall MIMO performance.

Chapter 5

Conclusion and Future Work

A compact self-isolated wideband MIMO antenna for 5G millimeter-wave applications has been successfully designed and analyzed using CST Studio Suite. The proposed antenna achieves intrinsic isolation through optimized radiator geometry, thereby eliminating the need for external decoupling structures and making it suitable for compact millimeter-wave devices. The incorporation of U-shaped, H-shaped, and circular slots on the radiating patches effectively suppresses surface currents and mutual coupling, resulting in wide impedance bandwidth over the 21–45 GHz range, high isolation, high radiation efficiency, and very low envelope correlation coefficient. Comprehensive simulation studies carried out on both 1×2 and 2×2 MIMO configurations confirm the robustness and effectiveness of the proposed design, demonstrating its strong potential for future fabrication and integration into high-performance 5G millimeter-wave communication systems.

5.1 Future Work

Although the proposed antenna shows promising performance through simulation, further work can be carried out to enhance and extend this study. Fabrication and experimental validation of the antenna can be performed to verify the simulated results. The design can be extended to larger MIMO arrays such as 4×4 or higher configurations to support advanced 5G and beyond-5G systems. Additionally, the antenna can be optimized for beam steering and integration with practical wireless devices. Exploration of alternative slot geometries, substrates, and reconfigurable techniques may further improve antenna performance and application scope.

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